CSE 471 — System Analysis and Design

Assignment 1

SDLC & DevOps · Feasibility Analysis · Requirements Determination

Total Marks: 30	Name: _____	ID: _____

Answer all questions. Show all working for numerical parts; PV factors are rounded to 3 d.p.

Section A — SDLC & DevOps (8 marks)

Question 1.[5]

(a) List the four phases of the SDLC and the key question each phase answers. (2)

(b) "QuickMeds", a pharmacy chain, is building an online prescription-refill system. Map each activity to its SDLC phase: (i) the sponsor prepares a feasibility study; (ii) analysts study the current refill workflow to gather requirements; (iii) the team designs the database schema and screen layouts; (iv) staff are trained and old records are converted. (2)

(c) Requirements are expected to change often after launch. Recommend one development methodology and justify in one sentence. (1)

Question 2.[3]

(a) State two problems with the traditional siloed Dev/Ops approach that motivated DevOps. (1)

(b) List the eight stages of the DevOps lifecycle in correct order. (2)

Section B — Feasibility Analysis (14 marks)

Question 3.[3]

(a) Name the three categories of feasibility and the central question each one answers. (1.5)

(b) Give one specific risk under each category for the QuickMeds project. (1.5)

Question 4 (Cost-Benefit Analysis).[8]

A system is evaluated over four years (figures in $). Development cost occurs only in Year 1; operational costs recur yearly. Discount rate = 10%, with Year 1 one period ahead. PV factors are given.

Item	Year 1	Year 2	Year 3	Year 4
Total Benefits	150,000	200,000	250,000	300,000
Development Costs	240,000	0	0	0
Operational Costs	60,000	60,000	70,000	80,000
PV factor @ 10%	0.909	0.826	0.751	0.683

(a) Compute PV of total benefits and total costs per year and the four-year totals. (3)

(b) Compute the NPV. (1)

(c) Compute the ROI = (PV Benefits − PV Costs) / PV Costs, as a percentage. (2)

(d) Determine the break-even point in years (cumulative PV net cash flow). (2)

Question 5.[3]

A university library wants an online study-room reservation system (currently done by phone/front desk); the Head Librarian is the champion.

(a) Write a System Request covering all five elements: Project Sponsor, Business Need, Business Requirements, Business Value, Special Issues/Constraints. (2)

(b) Classify each benefit as tangible or intangible: (i) 40% fewer double-bookings; (ii) improved student satisfaction; (iii) $8,000/year staff savings; (iv) better reputation. (1)

Section C — Requirements Determination (8 marks)

Question 6.[3]

(a) Distinguish functional from non-functional requirements in one sentence each. (1)

(b) Classify each statement as Functional, or Non-functional with its category (Operational / Performance / Security / Cultural & Political): (2)

#	Requirement statement
1	A student can search available rooms by date and time slot.
2	Search results must appear within 2 seconds.
3	Only authenticated students with a valid ID can reserve a room.
4	Personal data must comply with the national Data Protection Act.

Question 7.[2]

Choose the most appropriate elicitation technique (Interview / JAD / Questionnaire / Document Analysis / Observation) and justify briefly: (i) gather opinions fast from 500

dispersed customers; (ii) get four disagreeing departments to consensus in one structured workshop; (iii) check whether staff actually follow the order-entry process they describe.

Question 8 (Duration Analysis).[3]

A loan-approval process has the sequential steps below. Total elapsed time from submission to notification is 6 working days (8-hour days).

Process step	Processing time
Receive & log application	15 min
Verify customer documents	30 min
Perform credit check	20 min
Branch manager review	25 min
Notify customer of decision	10 min

(a) Compute total processing time and total elapsed time (minutes), and state what the gap indicates. (1.5)

(b) If “verify documents” and “credit check” run in parallel, compute the new processing time and the saving. (0.5)

(c) Name the analysis strategy best suited to: (i) small incremental user-suggested improvements; (ii) tracing several problems to shared root causes. (1)

— End of Assignment —

Submission Guidelines

1. You MUST write the answers on the paper. You can also use a Tab.

2. Make a cover page including your Assignment Number, Name, ID, Section, and Date.

3. Scan the pages and make a PDF. The file name should follow the format: StudentID_Section_Assignment_1 (eg 10301040_6_Assignment_1)

4. No late submissions will be accepted.

5. Upload the PDF to the Google Form. Use G-suite Mail only.